

Service Trade-off Analysis — STEM Career Path (AI Era)

Project: STEM Graduates Career Path — AI Era (Code For Good)

Doc type: Service-selection trade-off & cost justification — **for Code For Good Leadership / Board**

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Status: Audited draft for leadership review

Audience: Non-technical decision-makers (board, leadership, program leads)

Companion docs: [docs/Architecture-Design.md](#) · [docs/Customer-Journey.md](#) · [docs/Sitemap-and-Wireframes.md](#)

Pricing and program assumptions were re-verified in **June 2026** from vendor pages.

1. Executive summary

This platform is designed for a small 501(c)(3) nonprofit to launch a **pilot subscription/access-based STEM career program** with very low infrastructure cost, low volunteer maintenance, and strong accountability over who receives access.

The recommended setup is:

- **Frontend:** AWS Amplify Hosting, because Code For Good is already hosted on Amplify.
- **Backend:** AWS serverless services — Amazon Cognito, AWS Lambda, API Gateway, DynamoDB, SES, CloudWatch, and CloudTrail.
- **Donation / access payment:** Zeffy as the preferred low-cost path, with Stripe kept as the automation-heavy fallback.
- **Scheduling:** Cal.com or Calendly free tier, depending on whether the program needs multiple event types.
- **Security posture:** Launch lean with Cognito, API throttling, CloudWatch alarms, IAM least privilege, audit logs, and optional WAF later if abuse risk grows.

Executive Summary:

- **Infrastructure cost should be effectively \$0 during the pilot.** The realistic AWS gross cost for a small pilot is expected to be roughly **\$25–200/year without Amplify WAF**, or about **\$200–400/year if Amplify WAF is enabled**. Code For Good is aiming for **\$1,000 in AWS nonprofit credits (\$95 one time fee at signup/renewal per \$1,000. Max up to \$5,000)**, which should cover the expected pilot infrastructure cost with room to spare.
 - **The main real-money decision is payment processing.** Zeffy charges nonprofits **\$0 platform fees and \$0 processing fees**, funded by optional donor tips. Stripe has stronger payment automation and signed webhooks, but processing fees apply unless discounted nonprofit pricing is approved.
 - **Zeffy should be the default launch choice.**
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2. How to read this document

A trade-off analysis explains what we chose, what else we considered, and why the recommended option fits the organization. Each major service decision is evaluated through five lenses:

Lens	The question it answers
Specific charges	What does it actually cost us at pilot scale?
Ease of integration	How hard is it to connect to the rest of the platform?
Deployment	How easy is it to ship and update?
Maintenance	How much ongoing work does it create for volunteers?
Nonprofit fit	Does it protect donor funds, student data, and long-term sustainability?

3. Decision criteria & why they matter

A volunteer-supported nonprofit should not optimize the same way as a venture-funded startup. The priorities are:

1. **Low, predictable cost** — donor money should fund students and programming, not idle infrastructure.
2. **Low maintenance** — volunteers rotate; the stack must not require a full-time engineer.
3. **Accountability** — donor-funded access must be explainable and auditable.
4. **Data stewardship** — the platform will handle applicant and student information, including minors.
5. **No painful lock-in** — services should be replaceable if cost, policy, or scale changes.
6. **Sustainable volunteer onboarding** — future student developers should be able to understand and maintain the stack from public documentation.

These criteria drive the recommendation more than pure developer convenience.

4. Cost model: what actually drives cost

4.1 Cost drivers at pilot scale

The platform is mostly pay-per-use. Between cohorts, there should be little or no backend traffic, so cost stays low. The meaningful cost drivers are:

1. **Amplify Hosting usage** — usually low for a small static or React-style site.
2. **Backend API traffic** — Lambda + API Gateway, likely very low at pilot scale.
3. **Database and storage** — DynamoDB and S3, likely near-free at pilot scale.
4. **Email** — SES is cheap for transactional email.
5. **Security add-ons** — AWS WAF is the one line that can create a fixed monthly cost.
6. **Payment processing** — Zeffy is \$0 to the nonprofit; Stripe/PayPal style processors charge per transaction.

4.2 Illustrative annual run cost — recommended lean pilot

Assumptions: 501(c)(3), first pilot app, Code For Good already hosted on AWS Amplify, hundreds of users at most, monthly active users far below 10,000, basic transactional email, no paid WAF add-on at launch.

Line item	Basis	Estimated annual gross cost
Frontend hosting	AWS Amplify Hosting, light traffic/build usage	\$0–25
Identity	Amazon Cognito Essentials, under 10,000 MAU	\$0
Compute	AWS Lambda, low request volume	\$0–10
API	API Gateway HTTP API, low request volume	\$0–25
Database	DynamoDB, small table size and low read/write volume	\$0–15
Storage	S3 for receipts/log exports, small files	\$0–10
Email	Amazon SES at low transactional volume	\$1–10
Audit/monitoring	CloudWatch + CloudTrail, basic configuration	\$5–50
Scheduling	Cal.com or Calendly free tier	\$0
Secrets / EventBridge / SQS	Small usage	\$0–10
Subtotal — lean pilot	Without Amplify WAF	≈ \$25–200/year
AWS nonprofit credit target	Conservative planning assumption	–\$1,000/year
Expected net infrastructure cost	Pilot scale	\$0
Donation/access payment	Zeffy preferred; Stripe optional	Zeffy \$0; Stripe varies

Takeaway: even using a conservative **\$1,000 AWS credit target**, the expected pilot infrastructure cost should be covered. The biggest uncertainty is not normal AWS usage; it is whether the team enables fixed-cost security add-ons too early.

4.3 Optional security cost: Amplify WAF

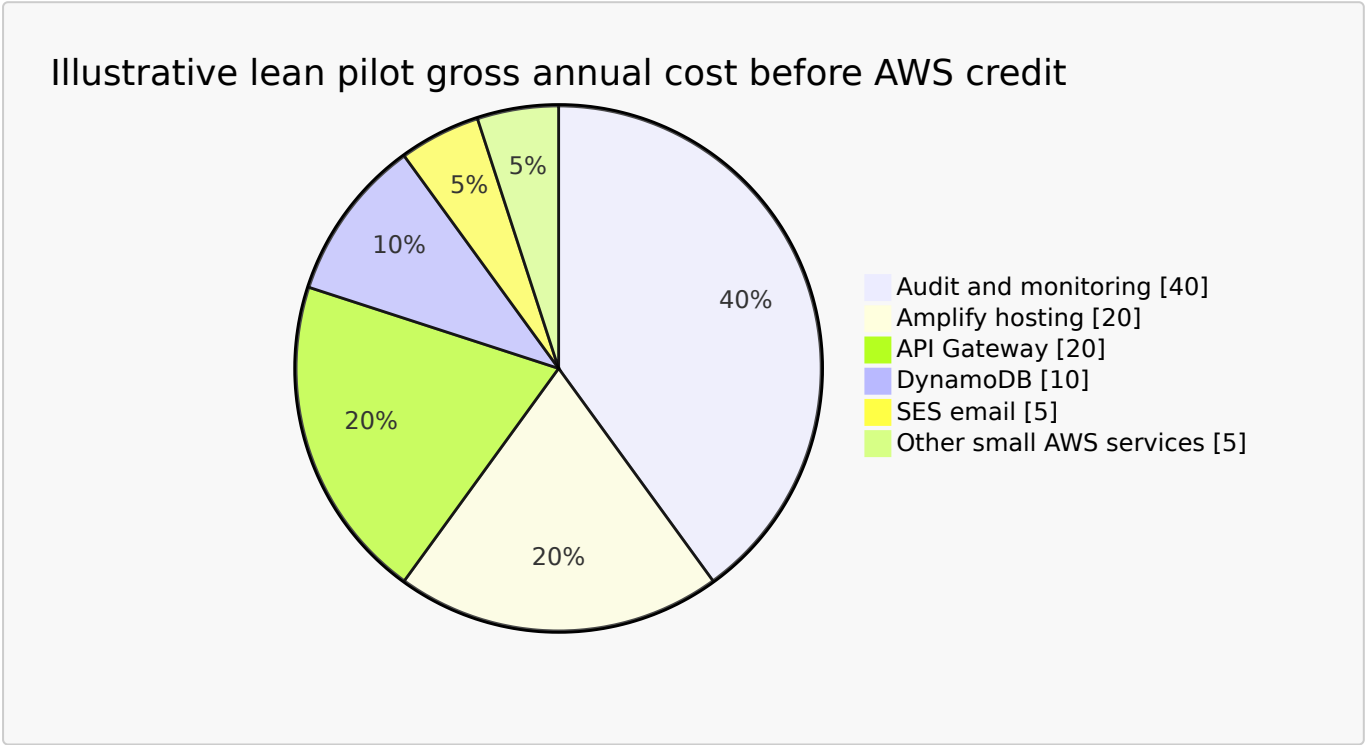
Amplify now supports direct AWS WAF integration, but it changes the cost picture. If WAF is attached directly to an Amplify app, AWS lists an **Amplify WAF integration charge of \$15/month per app**, plus normal AWS WAF charges.

Security option	Estimated cost	When to use
Lean launch security	\$0–low beyond normal logs/API usage	Best for pilot: Cognito, IAM, API throttling, CloudWatch alarms, CloudTrail logs
Amplify + WAF integration	~\$180/year + WAF usage	Use if the public site sees abuse, scraping, spam, or brute-force pressure

Security option	Estimated cost	When to use
S3 + CloudFront + WAF alternative	WAF usage without the Amplify integration fee	Consider later if WAF becomes necessary and hosting migration is acceptable

Recommendation: do **not** enable Amplify WAF by default on day one unless there is a known abuse risk. Launch lean, monitor traffic, then enable WAF when the evidence justifies it.

4.4 Cost share — lean pilot before credits



This chart intentionally excludes Amplify WAF. If WAF is enabled, it becomes the largest fixed infrastructure line item.

5. Nonprofit and compliance programs we rely on

Program / issue	Benefit or requirement	Why it matters
AWS Nonprofit Credit Program	Code For Good is a 501(c)(3) and is aiming for \$1,000 AWS credits	Conservative credit assumption still covers pilot infrastructure
AWS Free Tier / low usage pricing	Lambda, Cognito, DynamoDB, CloudFront/API-related services include free or low-cost usage bands	Keeps early usage near \$0, especially before scale
Zeffy	\$0 platform and payment-processing fees for nonprofits, funded by optional donor tips	Best fit for lowest-cost launch
Stripe nonprofit discount	Eligible nonprofits may receive discounted processing fees, subject to approval and region/card limits	Strong fallback if automation becomes more important than fee minimization

Program / issue	Benefit or requirement	Why it matters
IRS quid pro quo rules	If payment grants something of value, receipts may need disclosure language	Important because minimum contribution may unlock access
Minor/student privacy	High school users may be minors; under-13 users create COPPA risk	The platform should collect minimal data and set clear age/consent rules

Leadership action items:

1. Confirm the AWS nonprofit credit application path and timing.
2. Decide the minimum contribution threshold for automatic access.
3. Decide whether this is framed as a **donation, membership contribution, program access contribution, or sponsor-supported fee.**
4. Approve receipt language before launch.
5. Approve an age/guardian consent policy for high school participants.

6. Service-by-service trade-offs

6.1 Overall approach — AWS Amplify + AWS serverless vs. all-in-one platform vs. rented server

Option	Specific charges	Integration	Deployment	Maintenance	Nonprofit fit
AWS Amplify + serverless backend (chosen)	Expected \$0 net with \$1,000 credit; low gross pilot cost	Native AWS path; works with Cognito, Lambda, DynamoDB	Amplify makes frontend deploys easy	Low; no servers to patch	Best match because Code For Good already uses Amplify
Firebase / Supabase	Generous free tiers; may become opinionated at scale	Fastest to start	Very easy	Low, but data/auth/backend tied to vendor model	Good developer experience, weaker fit with existing AWS hosting
Render / Railway / Heroku / VPS	\$7–25+/month or more, often always-on	Simple mental model	Easy	Higher; app/runtime/server maintenance	Pays for idle time and weaker native audit story
Cloudflare stack	Very strong free/low-cost edge platform	Good for JS/edge apps	Easy for static and edge workloads	Low	Worth considering later, but switching

Option	Specific charges	Integration	Deployment	Maintenance	Nonprofit fit
					now adds churn

Verdict. Stay with AWS. Code For Good already uses Amplify, AWS nonprofit credits can offset pilot infrastructure, and the serverless backend avoids idle server cost. Switching clouds now would be architecture cosplay — fun for diagrams, bad for delivery.

6.2 Frontend hosting — AWS Amplify vs. S3 + CloudFront vs. Netlify / Vercel

Option	Cost profile	Strength	Weakness	Recommendation
AWS Amplify Hosting (chosen)	Low at pilot scale; paid WAF integration if enabled	Already used by Code For Good; easy deploys	Amplify WAF adds \$15/month per app + WAF charges	Use for launch
S3 + CloudFront	Very low static hosting cost	Cheapest AWS-native static hosting; strong WAF/CDN control	More setup and migration from current Amplify workflow	Keep as future cost-control option
Netlify	Easy deploys; free/paid tiers	Friendly developer workflow	Another vendor; build/runtime limits; less AWS audit cohesion	Fine, but not better here
Vercel	Excellent frontend DX	Great for Next.js	Hobby plan restrictions and paid per-seat model for org use	Avoid unless app specifically needs Vercel features

Verdict. Since the organization is already on Amplify, use Amplify Hosting for this pilot. Do not migrate to S3 + CloudFront unless WAF cost or hosting scale makes that worthwhile later.

6.3 Compute & API — Lambda + API Gateway vs. containers vs. VPS

- **Chosen:** AWS Lambda + API Gateway HTTP API.
- **Charges:** Lambda includes a large free usage band for small apps. API Gateway has a limited free-tier period and then low per-million-request pricing. At pilot scale, expected cost is near \$0 to a few dollars.
- **Alternatives:** ECS/Fargate, App Runner, EC2, Render, Railway, or VPS hosting.
- **Integration:** Lambda integrates cleanly with API Gateway, Cognito authorizers, DynamoDB, SES, EventBridge, SQS, CloudWatch, and CloudTrail.
- **Deployment:** deploy through AWS IaC or Amplify/backend CI/CD pipeline.

- **Maintenance:** no servers to patch, no container cluster to babysit.

Verdict. Lambda + API Gateway is the right pilot backend. Containers can wait until traffic is constant or the app needs long-running workloads.

6.4 Identity & sign-in — Amazon Cognito vs. Clerk / Auth0 / Firebase

Provider	Free / low-cost profile	Integration	Maintenance	Nonprofit fit
Amazon Cognito Essentials (chosen)	10,000 MAU free tier for direct/social users; low cost above that	Native AWS integration	Managed service	Best fit with Amplify/AWS backend
Clerk	Strong free tier and excellent developer experience	Easy frontend integration	Low	Good alternative, but another vendor and bill
Auth0	Polished enterprise auth	Strong enterprise features	Low	Overkill and likely more expensive later
Firebase Auth	Large free tier	Best inside Google ecosystem	Low	Good product, weaker fit with AWS-first stack

Verdict. Use Cognito Essentials. Do **not** enable Cognito Plus at launch unless threat-protection features are explicitly required, because Plus changes the cost profile. Keep the access model invite/admin-controlled rather than open public sign-up.

6.5 Database — DynamoDB vs. Postgres / Supabase

- **Chosen:** DynamoDB on-demand.
- **Charges:** DynamoDB includes a free storage allowance and low pay-per-request cost. Pilot usage should be near \$0.
- **Alternatives:** RDS/Aurora Postgres, Supabase, Neon, PlanetScale-style managed databases.
- **Integration:** DynamoDB works well with Lambda and event-driven access-state records.
- **Maintenance:** no database server, no patching, no backups to babysit at pilot scale.
- **Nonprofit fit:** good for low-cost, low-maintenance access records, applications, audit pointers, status transitions, and entitlement state.

Verdict. DynamoDB is the right pilot database if the data model is simple and access-pattern-driven. If the app later needs heavy relational reporting, move reporting data into Postgres or add a small analytics replica. Do not pay Postgres rent before the product earns it.

6.6 Email — Amazon SES vs. SendGrid / Mailgun / Postmark

- **Chosen:** Amazon SES.

- **Charges:** SES is priced around pennies per thousand emails. For approval notices, receipts, expiry reminders, and interview scheduling links, annual cost should be tiny.
- **Alternatives:** SendGrid, Mailgun, Postmark, Resend.
- **Integration:** SES integrates directly with Lambda and AWS IAM.
- **Maintenance:** main setup work is domain verification, SPF/DKIM/DMARC, and SES sandbox removal.

Verdict. Use SES for transactional email. Use a marketing/email platform only if the program later needs newsletters, audience segmentation, or donor campaigns.

6.7 Donation / access payment — Zeffy vs. Stripe vs. PayPal / Donorbox

This is the most important service decision because it affects both money and user access.

Option	Fee profile	Auto-grant access?	Integration	Best use case
Zeffy (recommended launch default)	\$0 platform fees and \$0 processing fees to the nonprofit	Possible, but less direct than Stripe; use API polling, Zapier, or admin review	Donation forms, donor data, read-only API, Zapier integrations	Cost-first nonprofit launch
Stripe Checkout	Standard or approved nonprofit-discounted processing fees	Excellent; signed webhooks can activate access immediately	Strong API/webhook ecosystem	Automation-first payment flow
PayPal Giving Fund / PayPal	Can be low-cost for eligible nonprofits	Limited / more awkward	Familiar to donors	Optional secondary donor preference
Donorbox	Platform fee plus processor fees	Some integrations	Easy donation widgets	Useful if donor-management UX matters more than cost

Recommended launch flow with Zeffy

1. User applies or signs up for the program.
2. App creates a pending account/application record in DynamoDB.
3. User is sent to a Zeffy form for the minimum contribution or recurring support option.
4. Backend verifies that a qualifying payment exists using one of these methods:
 - Zeffy API polling by email/payment reference,
 - Zeffy → Zapier → backend webhook,
 - admin review of Zeffy payment records.
5. If the contribution meets the minimum threshold, the backend marks the user as **ACTIVE** and adds the proper Cognito group/entitlement.
6. All status changes are logged for audit.

When to switch to Stripe

Use Stripe if leadership wants **near-real-time access activation** with fewer moving parts. Stripe webhooks are more mature for this exact product pattern. The trade-off is the processing fee.

Receipt and tax wording warning

If a minimum payment unlocks program access, the payment may not be a plain unrestricted donation. It may be partly a charitable contribution and partly a payment for benefits. Leadership should approve wording such as:

"Your contribution supports Code For Good's STEM Career Path program. If your contribution provides access to program services, the tax-deductible amount may be limited to the amount paid above the fair market value of benefits received. Please consult your tax advisor."

Verdict. Launch with Zeffy because the organization wants to preserve every dollar and is leaning that direction. Keep Stripe as the fallback if Zeffy automation becomes brittle.

6.8 Scheduling — Cal.com vs. Calendly vs. Google Calendar appointment scheduling

Option	Free-tier fit	Strength	Weakness	Recommendation
Cal.com	Free individual plan supports unlimited event types/calendars	Better if multiple interview types are needed	Another account/tool to manage	Strong default if free flexibility matters
Calendly	Free plan supports one event type	Very familiar and simple	Free tier is narrower	Good if one interview type is enough
Google Calendar appointment scheduling	Best if organization already uses Google Workspace	Keeps scheduling inside Google	Workspace/account dependency	Good if Code For Good already has Workspace

Verdict. Use **Cal.com** if the program needs separate booking links for high school, college, graduate, and young professional tracks. Use **Calendly** if there is only one standard 15-minute interview. Scheduling should stay external; do not build scheduling from scratch.

6.9 Security, rate limiting & audit

Launch security should be practical, not theatrical. The pilot does not need enterprise security cosplay, but it does need clean controls.

Recommended launch controls:

- Cognito authentication and role/group separation.

- No public admin routes.
- API Gateway throttling and request limits.
- IAM least privilege for Lambda Functions.
- DynamoDB conditional writes for idempotent access-state changes.
- CloudWatch alarms for error spikes and cost anomalies.
- CloudTrail enabled for account-level audit events.
- Manual override path for admins, with every override logged.
- No card data stored in Code For Good systems.

Optional later controls:

- Amplify WAF if public traffic shows abuse.
- Bot protection only if needed.
- More detailed CloudTrail data events for sensitive objects.
- Security review before expanding to larger cohorts.

Verdict. Launch lean. Add WAF only after risk justifies its fixed monthly cost.

7. Audience and data stewardship

The program audience includes:

- high school students,
- college students,
- graduate students,
- young professionals with 1–5 years of experience.

Because high school students may be minors, the platform should follow these rules:

1. **Do not target or knowingly collect data from children under 13.** If under-13 participation is ever considered, get legal review first.
2. **Add age-band collection, not exact birthdate unless needed.** Example: **Under 13, 13–17, 18+.**
3. **For minors, add parent/guardian consent language.** This can be lightweight for a pilot but should be explicit.
4. **Collect only necessary data.** Avoid sensitive demographic data unless it is truly needed for program reporting or grant requirements.
5. **Separate application data from payment data.** Zeffy/Stripe stores payment details; Code For Good stores only payment reference, status, amount threshold met, and audit metadata.
6. **Define retention.** Example: rejected applications retained 12 months, active member records retained while active plus 24 months, audit logs retained according to leadership policy.

Verdict. The mixed-age audience does not block launch, but it does require sharper privacy copy and consent handling than an adult-only app.

8. Vendor lock-in & exit strategy

Component	Lock-in level	Exit mitigation
Amplify Hosting	Low–Medium	Static frontend can move to S3/CloudFront, Netlify, Vercel, or Cloudflare Pages
Lambda/API Gateway	Medium	Keep business logic framework-light; isolate AWS event adapters
DynamoDB	Medium	Document access patterns; export data; migrate reporting to Postgres if needed
Cognito	Medium	Use email-based users and avoid unnecessary custom lock-in
Zeffy	Low	Store only payment references; Stripe/PayPal can replace it
Stripe fallback	Low	Payment processor is swappable if internal records remain processor-neutral
SES	Low	Email provider can be replaced with SendGrid/Postmark/Resend
Cal.com/Calendly	Low	Scheduling links are replaceable

Overall: the design is AWS-first, but not trapped. The key is to keep application logic clean and avoid scattering vendor-specific assumptions through the whole codebase.

9. Risks & mitigations

Risk	Likelihood	Impact	Mitigation
AWS credit is lower than expected or delayed	Medium	Low–Medium	Plan with conservative \$1,000 assumption; gross pilot cost remains manageable
Amplify WAF enabled too early	Medium	Medium	Delay WAF until needed; use API throttling and monitoring first
Zeffy automation is not real-time enough	Medium	Medium	Start with API polling/Zapier/admin review; switch access flow to Stripe if needed
Payment tied to access creates receipt/tax confusion	Medium	High	Use leadership-approved quid pro quo / benefit disclosure language
Minor participation creates privacy/consent risk	Medium	High	Age-band check, guardian consent for minors, no under-13 collection without review
AWS complexity overwhelms student volunteers	Medium	Medium	Use Amplify for frontend, serverless managed services, IaC, and clear runbooks
Cost spike from bug or abuse	Low–Medium	Medium	Budget alarms, API throttling, CloudWatch alerts, rate limits

Risk	Likelihood	Impact	Mitigation
Pricing changes after this document	Medium	Medium	Re-verify vendor pricing before build and before board approval
Vendor terms change	Low–Medium	Medium	Keep Zeffy/Stripe/payment abstraction and documented exit paths

10. Recommendation

1. **Approve AWS Amplify Hosting + AWS serverless backend** for the pilot.
2. **Apply for the \$1,000 AWS Nonprofit Credit** and use that as the conservative planning assumption.
3. **Launch Zeffy-first** for minimum contribution / subscription-style access because it preserves the full payment amount for the nonprofit.
4. **Implement automatic access carefully:** verify the Zeffy payment meets the minimum threshold before activating the Cognito entitlement.
5. **Keep Stripe as the fallback** if leadership later prioritizes instant, robust payment webhooks over zero processing fees.
6. **Do not enable Amplify WAF at launch** unless there is a known threat. Start with Cognito, API Gateway throttling, IAM least privilege, CloudWatch alarms, and CloudTrail.
7. **Add minor/privacy safeguards** before accepting high school applicants.
8. **Review receipt language** before launch because payment-triggered access may require quid pro quo disclosure.
9. **Reconfirm all pricing and terms before production.** The current numbers are planning estimates, not contracts.

11. One-page cost summary for the board

Category	Pilot recommendation	Estimated cost
Frontend	AWS Amplify Hosting	\$0–25/year at light usage
Backend	Lambda + API Gateway + DynamoDB + Cognito	\$0–50/year at pilot usage
Email	Amazon SES	\$1–10/year
Monitoring/audit	CloudWatch + CloudTrail	\$5–50/year
Optional WAF	Defer at launch	\$0 initially ; can add later
Scheduling	Cal.com or Calendly free tier	\$0
Payment processor	Zeffy preferred	\$0 to nonprofit
AWS nonprofit credit target	Conservative planning assumption	–\$1,000/year

Category	Pilot recommendation	Estimated cost
Expected net infrastructure cost	With credit	\$0

Plain-English board summary: Code For Good can pilot this platform on AWS with effectively no infrastructure cost after credits. The best launch path is Amplify + AWS serverless + Zeffy. The only decision that truly affects money and user experience is whether the organization wants Zeffy's zero-fee model or Stripe's stronger instant-payment automation.

12. References verified June 2026

- AWS Nonprofit Credit Program — <https://aws.amazon.com/government-education/nonprofits/nonprofit-credit-program/>
- AWS for Nonprofits — <https://aws.amazon.com/government-education/nonprofits/>
- AWS Amplify Hosting pricing — <https://aws.amazon.com/amplify/pricing/>
- AWS Amplify WAF pricing documentation — <https://docs.aws.amazon.com/amplify/latest/userguide/waf-pricing.html>
- AWS WAF pricing — <https://aws.amazon.com/waf/pricing/>
- Amazon Cognito pricing — <https://aws.amazon.com/cognito/pricing/>
- AWS Lambda pricing — <https://aws.amazon.com/lambda/pricing/>
- Amazon API Gateway pricing — <https://aws.amazon.com/api-gateway/pricing/>
- Amazon DynamoDB pricing — <https://aws.amazon.com/dynamodb/pricing/>
- Amazon SES pricing — <https://aws.amazon.com/ses/pricing/>
- Stripe nonprofit fee discount — <https://support.stripe.com/questions/fee-discount-for-nonprofit-organizations>
- Zeffy zero-fee fundraising — <https://www.zeffy.com/>
- Zeffy public API — <https://www.zeffy.com/integration/api>
- Zeffy integrations / Zapier — <https://support.zeffy.com/integrating-zeffy-with-other-sofware-74zts>
- Calendly pricing — <https://calendly.com/pricing>
- Cal.com pricing — <https://cal.com/pricing>
- FTC COPPA rule — <https://www.ftc.gov/legal-library/browse/rules/childrens-online-privacy-protection-rule-coppa>
- IRS quid pro quo contribution guidance — <https://www.irs.gov/charities-non-profits/substantiating-charitable-contributions>

Final pre-launch check: confirm AWS credit award amount, Zeffy automation method, Stripe fallback terms, minimum contribution amount, receipt language, age policy, and whether Amplify WAF is needed.